

CO5-AT2-Application Based Problem Solving

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1. Employee Management and Payroll Application using Java AWT and MySQL

MySQL Table Creation:

```
CREATE DATABASE employee_management;
USE employee_management;

CREATE TABLE department (
    department_id INT PRIMARY KEY AUTO_INCREMENT,
    department_name VARCHAR(50) NOT NULL UNIQUE
);

INSERT INTO department (department_name)
VALUES
('HR'),
('IT'),
('Finance'),
('Marketing'),
('Sales'),
('Operations');
```

```
CREATE TABLE employee (
    employee_id INT PRIMARY KEY,
    name VARCHAR(50) NOT NULL,
    gender VARCHAR(10) NOT NULL,
    dob DATE NOT NULL,
    contact VARCHAR(15) NOT NULL,
    email VARCHAR(100) NOT NULL UNIQUE,
    address VARCHAR(200),
    department_id INT NOT NULL,
    designation VARCHAR(50) NOT NULL,
    date_of_joining DATE NOT NULL,
    basic_salary DECIMAL(10,2) NOT NULL,
    allowances DECIMAL(10,2) DEFAULT 0,
    deductions DECIMAL(10,2) DEFAULT 0,
    employment_status VARCHAR(20) NOT NULL,

    FOREIGN KEY (department_id)
        REFERENCES department(department_id),

    CHECK (basic_salary >= 0),
    CHECK (allowances >= 0),
    CHECK (deductions >= 0)
);
```

```
CREATE TABLE payroll (
    payroll_id INT PRIMARY KEY AUTO_INCREMENT,
    employee_id INT NOT NULL,
    basic_salary DECIMAL(10,2) NOT NULL,
    allowances DECIMAL(10,2) DEFAULT 0,
    deductions DECIMAL(10,2) DEFAULT 0,
    gross_salary DECIMAL(10,2) NOT NULL,
    net_salary DECIMAL(10,2) NOT NULL,
    payroll_date DATE NOT NULL,

    FOREIGN KEY (employee_id)
        REFERENCES employee(employee_id)
);
```

69 ● SHOW TABLES;

Result Grid		Filter Rows:
	Tables_in_employee_management	
▶	department	
	employee	
	payroll	

71 ● `SELECT * FROM department;`

Result Grid   Filter Rows: Edit

	department_id	department_name
▶	3	Finance
	1	HR
	2	IT
	4	Marketing
	6	Operations
	5	Sales
✱	NULL	NULL

DBConnection.java

```

1 package employee;
2 import java.sql.Connection;
3 import java.sql.DriverManager;
4 public class DBConnection {
5     public static Connection getConnection() {
6         Connection con = null;
7         try{
8             Class.forName("com.mysql.cj.jdbc.Driver");
9             con = DriverManager.getConnection(
10                 "jdbc:mysql://localhost:3306/employee_management",
11                 "root",
12                 "root"
13             );
14             System.out.println("Database Connected Successfully!");
15         }
16         catch (Exception e){
17             e.printStackTrace();
18         }
19         return con;
20     }
21     public static void main(String[] args){
22         Connection con = getConnection();
23         if (con != null) {
24             System.out.println(
25                 "Connection is working!"
26             );
27         }
28         else {
29             System.out.println(
30                 "Connection failed!"
31             );
32         }
33     }
34 }

```

Employee.java

```

1 package employee;
2 import java.sql.Date;
3 public class Employee{
4     private int employeeId;
5     private String name;
6     private String gender;
7     private Date dateOfBirth;
8     private String contactNumber;
9     private String email;
10    private String address;
11    private String department;
12    private String designation;
13    private Date dateOfJoining;
14    private double basicSalary;
15    private double allowances;
16    private double deductions;
17    private String employmentStatus;
18    public Employee(){
19    }
20    public Employee(int employeeId, String name, String gender,
21        Date dateOfBirth, String contactNumber,
22        String email, String address,
23        String department, String designation,
24        Date dateOfJoining, double basicSalary,
25        double allowances, double deductions,
26        String employmentStatus) {
27        this.employeeId = employeeId;
28        this.name = name;
29        this.gender = gender;
30        this.dateOfBirth = dateOfBirth;
31        this.contactNumber = contactNumber;
32        this.email = email;
33        this.address = address;
34        this.department = department;
35        this.designation = designation;

```

EmployeeDAO.java

```
package employee;
import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.Date;
import java.util.ArrayList;
import java.util.List;
public class EmployeeDAO{
    public boolean addEmployee(Employee emp){
        String sql =
            "INSERT INTO employee "
            + "(employee_id, name, gender, dob, contact, email, address, "
            + "department_id, designation, date_of_joining, basic_salary, "
            + "allowances, deductions, employment_status) "
            + "VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?)";
        try (Connection con = DBConnection.getConnection();
            PreparedStatement ps = con.prepareStatement(sql)){
            int departmentId =
                getDepartmentId(con, emp.getDepartment());
            if (departmentId == -1){
                System.out.println("Department not found.");
                return false;
            }
            ps.setInt(1, emp.getEmployeeId());
            ps.setString(2, emp.getName());
            ps.setString(3, emp.getGender());
            ps.setDate(4, emp.getDateOfBirth());
            ps.setString(5, emp.getContactNumber());
            ps.setString(6, emp.getEmail());
            ps.setString(7, emp.getAddress());
            ps.setInt(8, departmentId);
            ps.setString(9, emp.getDesignation());
            ps.setDate(10, emp.getDateOfJoining());
            ps.setDouble(11, emp.getBasicSalary());
            ps.setDouble(12, emp.getAllowances());
            ps.setDouble(13, emp.getDeductions());
            ps.setString(14, emp.getEmploymentStatus());
            int rows = ps.executeUpdate();
            return rows > 0;
        } catch (Exception e){
            e.printStackTrace();
            return false;
        }
    }
    public boolean employeeExists(int employeeId){
        String sql =
            "SELECT employee_id "
            + "FROM employee "
            + "WHERE employee_id = ?";
        try (Connection con = DBConnection.getConnection();
            PreparedStatement ps = con.prepareStatement(sql)){
            ps.setInt(1, employeeId);
            ResultSet rs = ps.executeQuery();
            return rs.next();
        } catch (Exception e){
            e.printStackTrace();
            return false;
        }
    }
    public Employee getEmployee(int employeeId){
        String sql =
```

EmployeeManagementSystem.java

```
package employee;
import java.awt.Button;
import java.awt.Choice;
import java.awt.Color;
import java.awt.Font;
import java.awt.Frame;
import java.awt.GridLayout;
import java.awt.Label;
import java.awt.Panel;
import java.awt.TextArea;
import java.awt.TextField;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.awt.event.WindowAdapter;
import java.awt.event.WindowEvent;
import java.sql.Date;
import java.util.List;
public class EmployeeManagementSystem extends Frame
    implements ActionListener{
    Label titleLabel;
    Label idLabel;
    Label nameLabel;
    Label genderLabel;
    Label dobLabel;
    Label contactLabel;
    Label emailLabel;
    Label addressLabel;
    Label departmentLabel;
    Label designationLabel;
    Label joiningLabel;
    Label basicSalaryLabel;
    Label allowancesLabel;
    Label deductionsLabel;
    Label statusLabel;
    TextField idText;
    TextField nameText;
    TextField dobText;
    TextField contactText;
    TextField emailText;
    TextField designationText;
    TextField joiningText;
    TextField basicSalaryText;
    TextField allowancesText;
    TextField deductionsText;
    Choice genderChoice;
    Choice departmentChoice;
    Choice statusChoice;
    TextArea addressText;
    Button addButton;
    Button viewButton;
    Button searchButton;
    Button updateButton;
    Button deleteButton;
    Button clearButton;
    TextArea outputArea;
    Label payrollTitle;
    Label grossLabel;
    Label netLabel;
    Button calculatePayroll;
    EmployeeDAO dao;
    public EmployeeManagementSystem() {
```

Payroll.java

```
package employee;
public class Payroll{
    private double basicSalary;
    private double allowances;
    private double deductions;
    public Payroll(double basicSalary,
        double allowances,
        double deductions){
        this.basicSalary = basicSalary;
        this.allowances = allowances;
        this.deductions = deductions;
    }
    public double calculateGrossSalary(){
        return basicSalary + allowances;
    }
    public double calculateNetSalary(){
        return calculateGrossSalary()-deductions;
    }
    public double getBasicSalary(){
        return basicSalary;
    }
    public double getAllowances(){
        return allowances;
    }
    public double getDeductions(){
        return deductions;
    }
}
```

Employee and payroll Application:

Employee Management and Payroll System

EMPLOYEE MANAGEMENT AND PAYROLL SYSTEM

Employee ID

Department

HR

Name

Designation

Gender

Male

Date of Joining

Date of Birth

Basic Salary

Contact Number

Allowances

Email

Deductions

Address

Employment Status

Active

ADD

VIEW

SEARCH

UPDATE

DELETE

CLEAR

No employee records found.

PAYROLL CALCULATION

Gross Salary :

Net Salary :

CALCULATE PAYROLL

Adding the Employee Details:

Employee Management and Payroll System

EMPLOYEE MANAGEMENT AND PAYROLL SYSTEM

Employee ID

107

Department

Operations

Name

Virat

Designation

Leader

Gender

Male

Date of Joining

2017-02-03

Date of Birth

1988-10-10

Basic Salary

70000

Contact Number

9121234567

Allowances

4500

Email

virat@gmail.com

Deductions

2000

Address

Delhi

Employment Status

Active

ADD

VIEW

SEARCH

UPDATE

DELETE

CLEAR

@gmail.com	Kanchipuram	IT	Software Developer	2024-06-01	50000.00	5000.00
nesg@gmail.com	Vellore	Marketing	Sales	2020-07-02	40000.00	500.00
nn@gmail.com	Coimbatore	Operations	Commading	2019-01-01	34000.00	1200.00
iya@gmail.com	Melmaruvathur	Finance	Collection Team	2017-04-04	57000.00	4300.00
ar@gmail.com	Thanjavur	Sales	Selling Team	2015-08-07	60000.00	7000.00
wer@gmail.com	Thiruvannamalai	Finance	Collection Team	2024-05-06	54000.00	3400.00

PAYROLL CALCULATION

Gross Salary :

Net Salary :

CALCULATE PAYROLL

View the Employee Details:

Employee Management and Payroll System

EMPLOYEE MANAGEMENT AND PAYROLL SYSTEM

Employee ID

Department

HR

Name

Designation

Gender

Male

Date of Joining

Date of Birth

Basic Salary

Contact Number

Allowances

Email

Deductions

Address

Employment Status

Active

ADD

VIEW

SEARCH

UPDATE

DELETE

CLEAR

102	Ganesh	Male	2000-01-01	9111111122	ganesh@gmail.com	Vellore	Marketing
103	John	Male	1999-08-02	9111111177	john@gmail.com	Coimbatore	Operator
104	Priya	Female	1995-09-08	9123456789	priya@gmail.com	Melmaruvathur	Finance
105	Galaxy Star	Male	1990-03-02	9111111118	star@gmail.com	Thanjavur	Sales
106	Power star	Male	2000-01-03	9121212123	power@gmail.com	Thiruvannamalai	Finance
107	Virat	Male	1988-10-10	9121234567	virat@gmail.com	Delhi	Operator

PAYROLL CALCULATION

Gross Salary :

Net Salary :

CALCULATE PAYROLL

Search the Employee Details:

Search Employee

Search By: Employee ID

Search Value:

SEARCH

CANCEL

Search Employee

Search By: Employee ID

Search Value: 101

SEARCH

CANCEL

Employee Management and Payroll System

EMPLOYEE MANAGEMENT AND PAYROLL SYSTEM

Employee ID

101

Department

IT

Name

Dhilip.K

Designation

Software Developer

Gender

Male

Date of Joining

2024-06-01

Date of Birth

2006-08-29

Basic Salary

50000.0

Contact Number

9111111111

Allowances

5000.0

Email

dk@gmail.com

Deductions

2000.0

Address

Kanchipuram

Employment Status

Active

ADD

VIEW

SEARCH

UPDATE

DELETE

CLEAR

Employee found successfully.

PAYROLL CALCULATION

Gross Salary :

Net Salary :

CALCULATE PAYROLL

Update the Employee Details:

I am updating the John Address Coimbatore into Kolkata

Employee Management and Payroll System

EMPLOYEE MANAGEMENT AND PAYROLL SYSTEM

Employee ID	103	Department	Operations
Name	John	Designation	Commading
Gender	Male	Date of Joining	2019-01-01
Date of Birth	1999-08-02	Basic Salary	34000.0
Contact Number	9111111177	Allowances	1200.0
Email	john@gmail.com	Deductions	500.0
Address	Kolkata	Employment Status	Active

Employee updated successfully.

PAYROLL CALCULATION

Gross Salary : Net Salary :

73 ● `SELECT *FROM employee;`

74

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	employee_id	name	gender	dob	contact	email	address	department_id	designation
▶	101	Dhilip.K	Male	2006-08-29	9111111111	dk@gmail.com	Kanchipuram	2	Software Dev
	102	Ganesh	Male	2000-01-01	9111111122	ganesg@gmail.com	Vellore	4	Sales
	103	John	Male	1999-08-02	9111111177	john@gmail.com	Kolkata	6	Commading
	104	Priya	Female	1995-09-08	9123456789	priya@gmail.com	Melmaruvathur	3	Collection Te
	105	Galaxy Star	Male	1990-03-02	9111111118	star@gmail.com	Thanjavur	5	Selling Team
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Address Updated Successfully.

103	John	Male	1999-08-02	9111111177	john@gmail.com	Kolkata	6	Commading
-----	------	------	------------	------------	----------------	---------	---	-----------

Delete Employee Details:

Employee ID

I am deleting the 104 Employee Details.

73 • `SELECT *FROM employee;`

74

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	employee_id	name	gender	dob	contact	email	address	department_id	designation
▶	101	Dhilip.K	Male	2006-08-29	9111111111	dk@gmail.com	Kanchipuram	2	Software Devel
	102	Ganesh	Male	2000-01-01	9111111122	ganesg@gmail.com	Vellore	4	Sales
	103	John	Male	1999-08-02	9111111177	john@gmail.com	Kolkata	6	Commading
	105	Galaxy Star	Male	1990-03-02	9111111118	star@gmail.com	Thanjavur	5	Selling Team
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Clear:

Before clicking the clear button:

Employee Management and Payroll System

EMPLOYEE MANAGEMENT AND PAYROLL SYSTEM

Employee ID
101

Department
IT

Name
Dhilip.K

Designation
Software Developer

Gender
Male

Date of Joining
2024-06-01

Date of Birth
2006-08-29

Basic Salary
50000.0

Contact Number
9111111111

Allowances
5000.0

Email
dk@gmail.com

Deductions
2000.0

Address
Kanchipuram

Employment Status
Active

ADD
VIEW
SEARCH
UPDATE
DELETE
CLEAR

Employee found successfully.

PAYROLL CALCULATION

Gross Salary :
Net Salary :

CALCULATE PAYROLL

After clicking the clear button:

Employee Management and Payroll System

EMPLOYEE MANAGEMENT AND PAYROLL SYSTEM

Employee ID		Department	HR
Name		Designation	
Gender	Male	Date of Joining	
Date of Birth		Basic Salary	
Contact Number		Allowances	
Email		Deductions	
Address		Employment Status	Active

ADD

VIEW

SEARCH

UPDATE

DELETE

CLEAR

Employee found successfully.

PAYROLL CALCULATION

Gross Salary :
Net Salary :

CALCULATE PAYROLL

Payroll Calculation:

Employee Management and Payroll System

EMPLOYEE MANAGEMENT AND PAYROLL SYSTEM

Employee ID	102	Department	Marketing
Name	Ganesh	Designation	Sales
Gender	Male	Date of Joining	2020-07-02
Date of Birth	2000-01-01	Basic Salary	40000.0
Contact Number	9111111122	Allowances	500.0
Email	ganesg@gmail.com	Deductions	200.0
Address	Vellore	Employment Status	Active

ADD

VIEW

SEARCH

UPDATE

DELETE

CLEAR

Employee found successfully.

PAYROLL CALCULATION

Gross Salary : 40500.00
Net Salary : 40300.00

CALCULATE PAYROLL

MySQL Database Output:

[illegible]